

PROFESSIONAL SUMMARY

Machine Learning Systems Engineer with over 5 years of experience developing resilient distributed systems and petabyte-scale AI pipelines for LLM pre-training and post-training with strong expertise in architecting data platforms and ML systems. Experienced in building resilient services under demanding SLOs, optimizing infrastructure, debugging production systems and driving cross-team architecture improvements.

EXPERIENCE

AI/ML Data Engineer, Amazon AGI

Mar 2025 – Present

- Engineered **petabyte-scale** pipelines orchestrating data processing of **multi-modal (text, audio, image, video)** datasets with **PySpark**, **FFmpeg**, and **AWS** services (S3, EMR, Glue, Step Functions), producing training corpora for **4+ LLM pre-training** runs across the Amazon Nova models.
- Productionized the **video-eval** harness for Nova ablation studies, implementing **LLM-as-judge** scoring and **ML classification** filters over **100,000+** hours of licensed video, saving **\$4.86M** in acquisition costs and **2+** weeks of manual engineering effort per ablation cycle.
- Designed deterministic partitioning strategy, optimizing **distributed** pipeline I/O throughput eliminating data skew and cutting processing latency by **70%** (p95 reduced from 4 hours to 72 mins) unblocking training deadlines for AGI science teams.
- Integrated **multimodal** compute engine to enrich metadata and enforcing AI safety guardrails with CSAM detection, PII anonymization, toxicity filtering, Speech-to-Text (STT) diarization and waveform extraction to generate highly dense, cross-modal feature engineering.
- Built **MCP** servers, custom **agent harness** (planner/executor loop, tool registry, eval harness) indexing **10+** sources (agent traces, code repos, wikis, runbooks, application logs) and adopted across 4+ teams, reducing debugging cycles by 80%.

Data Engineer, Coinbase

Oct 2024 – March 2025

- Architected multi-engine orchestration layer with **Airflow** to schedule/hand-off ML workloads across **4+** tools and built key operators, migrating **200+** legacy dags and improving feature freshness and model accuracy scores by 15%.

ML Platform Engineer, Apple

Dec 2023 – Sep 2024

- Shipped features on the **foundation-model** inference and training platform, productionizing batch inference servers on **Ray** endpoints, accelerating distributed DeepSpeed **PyTorch** training and engineering stage-level **GPU** scheduling with profile observability that eliminated idle GPU spend by 15% across clusters.
- Implemented multi-modal **feature transformations** with **Iceberg**, **Spark** and **Flink** jobs to register, store, discover and reuse features across ML workloads and tools (MLflow), serving **100+** feature sets with reduced data movement and across teams.

Software Engineer- Data, Wipro

Jun 2019 – Dec 2021

- Designed **HIPAA** compliant Medallion architecture with auto-scaling **StepFunctions** for PII anonymization and built CI/CD pipeline using **AWS Sagemaker** reducing **NLP** model (SpaCy NER) deployment by 70% for batch data loads.
 - Implemented **feature engineering** for predictive models on healthcare use-cases (re-admission risk, staff scheduling, revenue assurance, waste) with Jupyter notebooks and libraries (**Pandas**, **NumPy**, **pydicom**, **Scikit-learn**) building high-quality datasets for academic researchers.
 - Deployed distributed **SQL query engine** (Dremio) on Kubernetes, optimizing cross-platform queries across S3 and Snowflake and improving query performance with caching, indexing and partitioning by 60% across teams.
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PROJECTS

Simulation of Distributed Machine Learning Platforms [PyTorch, OpenAI Gym, MuJoCo, YOLOv8, Websockets]

[Github](#)

- Architected a Reinforcement Learning pipeline to train autonomous Sphero robots for Robo-Cup soccer, successfully transferring policies trained in a virtual environment directly to real-world hardware by integrating with YOLOv8 computer vision model.
- Implemented Deep Deterministic Policy Gradient (DDPG) algorithm from scratch in PyTorch, utilizing separate Actor-Critic networks and experience replay buffers to achieve stable convergence in continuous control tasks.

Terminal-Task-Gen: LLM Terminal Agents [Pydantic, Qwen3/Llama, Hugging Face TRL, Docker]

[Github](#)

- Implemented a two-stage synthetic data generation pipeline inspired by NVIDIA's Nemotron-Terminal (arXiv:2602.21193), converting real math, code, and SWE-Bench problems into terminal agent training tasks using a seed-based LLM task adaptation system with auto-generated pytest verification suites.

Parameter-Golf-LM [PyTorch, Hugging Face, FineWeb]

[Hugging Face](#)

- Built a corpus engineering pipeline for OpenAI's Parameter Golf challenge, blending FineWeb-edu, StarCoder and OpenWebMath under 8B-token budget, applying cross-source MinHash-LSH deduplication and retokenizing with the baseline SentencePiece vocabulary
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SKILLS SUMMARY

Programming Languages: Python, C++, CUDA, JavaScript

AI/ML: PyTorch, TensorFlow, JAX, Hugging Face, llama.cpp, vLLM, PEFT, Evals, Ray, SageMaker, Collab, FAISS

Tools/ Frameworks: Pydantic, Spark, Kafka, Docker, Kubernetes, Postgres, Redis, Tableau [\[DASHBOARD\]](#)

Certifications: Apache Airflow Champion [link](#), AWS Solution Architect [link](#)

Current Focus: Attention mechanisms, Data Parallelism, Synthetic Data Generation, Agent skills, Inference Engineering, Distillation, Quantization

EDUCATION

San Jose State University, MS in Software Engineering [GPA: 3.7/4.0]

Jan 2022 – Dec 2023

Relevant Coursework: Enterprise Distributed Systems, Advanced Data Mining, Machine Learning

VMEG, B.Tech in Computer Science [GPA: 3.8/4.0]

Aug 2015 – May 2019

Relevant Coursework: Data Structures, Data Mining, Operating Systems, Design Patterns, Adv Mathematics